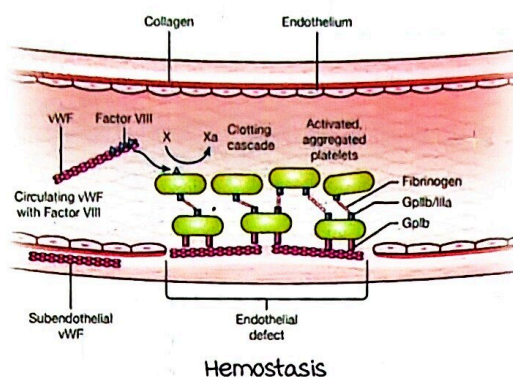


BLEEDING DISORDERS

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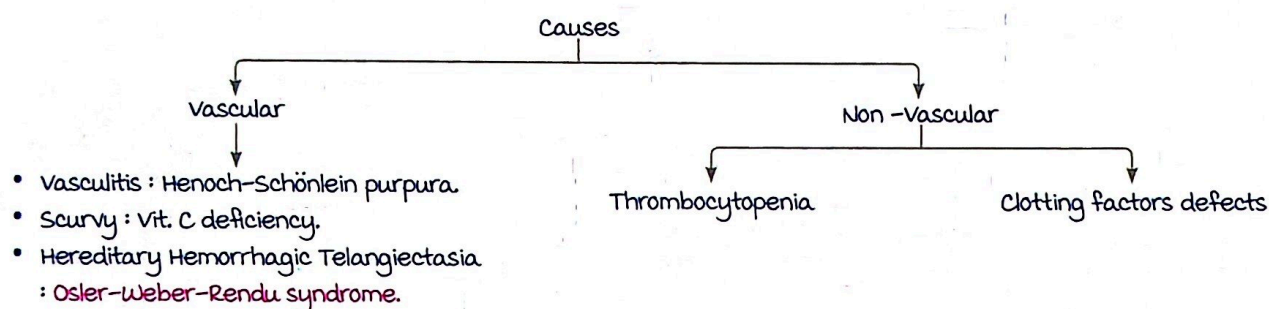
Normal hemostasis depends on

- Blood vessel
- Platelets
- Clotting factors



Etiology

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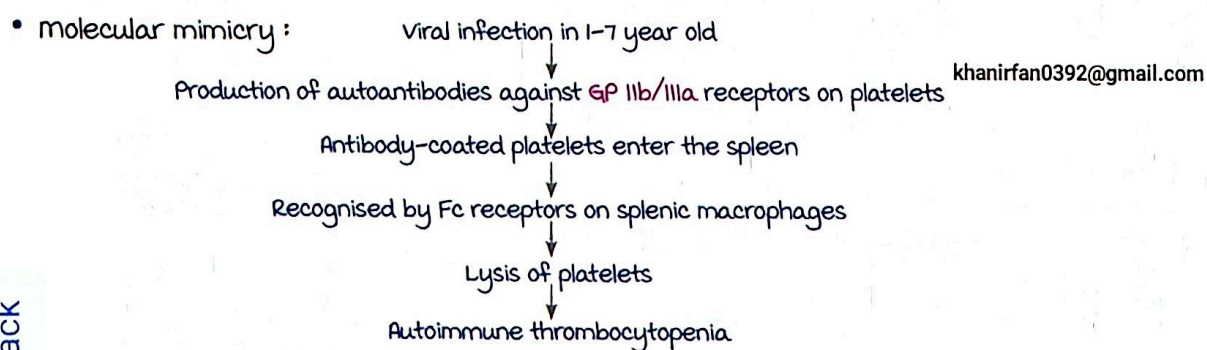
Differences b/w Platelet Disorders & Coagulation Disorders :

	Platelet disorders	Coagulation disorders
Site of bleeding	mucocutaneous	Deep : muscles & joints (Hemarthrosis)
Petechiae	Present	Absent
Ecchymoses	multiple & small	Single & large
Bleeding after minor trauma	Present	Absent
Bleeding after surgery (Delayed)	Absent	Present

Immune Thrombocytopenic Purpura (ITP)

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PATHOGENESIS :



Compensatory changes in bone marrow : ↑ megakaryocyte production.

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TYPES :

Acute (<3 months)

Persistent (3-12 months)

Chronic (>12 months)

- m/c (60-75%).
- Spontaneous recovery seen.

CLINICAL FEATURES OF ACUTE ITP :

- Age : 1-7 yrs.
- Viral infection : $\xrightarrow{1-4 \text{ weeks}}$ Sudden onset of mucocutaneous bleeds (Petechiae >> epistaxis, oral oozing).
- Child appears well.
- Rarely :
 - GI bleeds.
 - Hematuria.
 - CNS bleeds (Fatal).
- Lymphadenopathy & splenomegaly are **not** seen in acute ITP.
 - If present, evaluate for another etiology (Eg : leukemia).

INVESTIGATIONS :

1. Blood Count :

Thrombocytopenia

 $\xrightarrow[\text{bone marrow}]{\text{Compensation by}}$

- 2. Peripheral smear : Normal sized & large (Young platelets).
- 3. Bone marrow aspiration : $\uparrow$ megakaryocytes.

MANAGEMENT :

1. Supportive (majority of cases are self-limiting).
2. Decision to treat is :

- **Not based on platelet counts.**
- Based on clinical symptoms (Organ bleeds/ $\uparrow$ number of bleeds).

Treatment options :

- IV immunoglobulins (**Best**) $\rightarrow \downarrow$ macrophage activity $\rightarrow \downarrow$ platelet destruction in spleen. khanirfan0392@gmail.com
- Steroids (Prednisolone) : $\uparrow$ platelet production.
- Platelet transfusion : Only used for **life-threatening bleeds.**

Note :

Other causes of thrombocytopenia :

- Aplastic anemia.
- marrow infiltration : Leukemia/storage disorders.
- Connective tissue disorders (Eg : SLE).
- Infections : **Parvovirus**, **EBV**, Varicella, Rubella, mumps.

• **Drugs :**

- Heparin.
- Valproate, Phenytoin.
- NSAIDs.
- Penicillins & Cephalosporins.

Inherited Bleeding Disorders

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HEMOPHILIA :

Clotting factor defect → Delayed clot formation → Bleeding manifestations.

TYPES :

Type	Clotting factor deficiency	Mode of inheritance
Hemophilia A (m/c)	Factor VIII	X-linked recessive (males are affected more)
Hemophilia B	Factor IX	
Hemophilia C	Factor XI	Autosomal recessive

Clinical Features :

- Hemarthrosis : Bleeding into a joint.
- **Hallmark feature** of hemophilia :
- m/c joint affected : **Knee** > elbow, ankle
- Target joint : Repeated bleeding into same joint.

↓
Fibrosis

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Hemarthrosis of knee joint

Restriction of joint movements (m/c cause of morbidity).

Classification :

Severity	Clotting Factor Activity	Spontaneous Bleeds	Features
mild	5-40%	Absent	Unnoticed until adolescence
moderate	1-4%	mostly absent	Bleeding after trauma or surgery
Severe	<1%	Present	• Hemarthrosis, muscle Hematoma. • Rare bleeds : • CNS bleeds • Iliopsoas bleed → Shock • Neonatal : • Hematoma following I.M injection, circumcision. • Large cephalhematoma.

Investigations :

1. Coagulation Profile : ↑APTT (Intrinsic pathway compromise).
2. Factor Assay : **Best** test.
 - Estimates level of clotting factors in blood.

management :

Factor replacement therapy :

Dose calculation (based on desired % increase):

- Hemarthrosis : 40%.
- CNS bleeds : 100%.

Dose = % desired increase × body weight of child × K.

where **K : Constant**

- 0.5 : Hemophilia A.
- 1.4 : Hemophilia B.

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Other treatment options :

- Fresh Frozen Plasma (FFP).
- Cryoprecipitate : Contains Factor VIII & fibrinogen.
 - useful only in Hemophilia A.

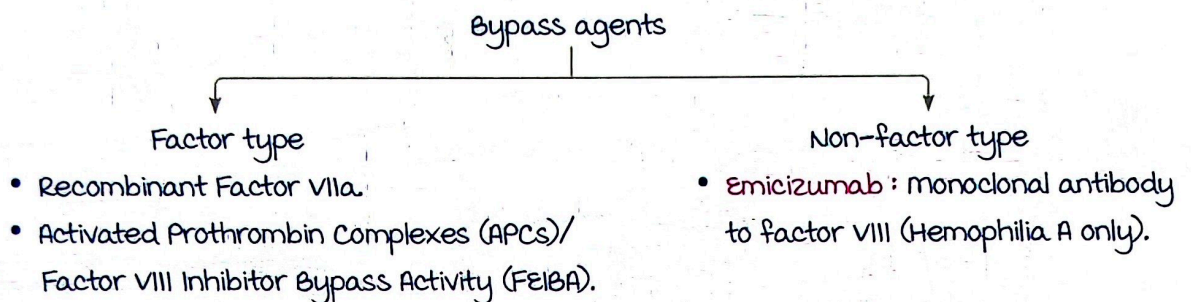
Complication :

Antibodies to clotting factor :

- most serious complication of therapy.
- Patients stop responding to factor replacement.

Treatment :

Bypass agents → Bypass the defective clotting factor in the cascade.

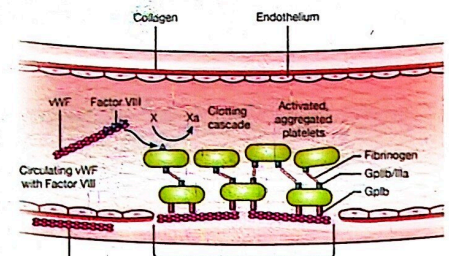


VON WILLEBRAND DISEASE (VWD) :

- m/c inherited bleeding disorder worldwide.
- most commonly : Autosomal dominant inheritance.

Functions of von Willebrand Factor (vWF) :

- Binds to Factor VIII → ↑ VIII half-life.
- Acts as a bridge b/w platelets & subendothelial collagen.



Functions of vWF

Types :

Type 1	• m/c (80% cases). • Quantitative defect. • Partial deficiency of vWF.
Type 2	• Qualitative defect. • Dysproteinemia : Function of vWF is abnormal, levels are normal.
Type 3	• most severe. • Quantitative defect. • Complete absence of vWF → Resembles Factor VIII deficiency (Hemophilia A).

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Clinical features :

- mucocutaneous bleeds (epistaxis, gum bleeds).
- **menorrhagia** in females.
- Severe form resembles Factor VIII deficiency (Hemophilia A).

Investigation :

1. Coagulation profile : $\uparrow$ APTT (Factor VIII activity affected).
 2. vWF antigen levels : used in Type 1 & Type 3 vWD.
 3. vWF multimer testing : Tests protein structure & functioning.
 4. vWF ristocetin cofactor assay :
 - Ristocetin induces binding of vWF to platelets.
 - vWD : $\downarrow$ binding of vWF to platelets.
- } used in Type 2 vWD

management :

- Desmopressin : Induces release of stored vWF from endothelial cells.
 - useful in **Type 1 vWD**.
- vWF-containing concentrates :
 - Plasma-derived
 - Recombinant
 - useful in Type 2 & Type 3 vWD.